

Amazon Web Services (AWS)

A Case Study

Outline

- 1 Understanding Cloud Computing
- 2 Benefits of AWS
- 3 Types of Cloud Services
- 4 Comparison: Best Public Cloud Providers
- 5 AWS Use Cases
- 6 Related Case-Studies - Netflix, Siemens, Adobe, Hotstar, BMW

Understanding Cloud Computing

Cloud Computing

- Cloud computing is the **on-demand availability of computer system resources, especially data storage (cloud storage) and computing power**, without direct active management by the user.
- Many times, cloud computing is **used to describe data centres available to many users over the Internet**.
- Large clouds, predominant today, often have functions distributed over multiple locations from central servers.



Amazon Web Services (AWS)

Among many Cloud Platforms in the market, AWS is one of the most popular clouds.

- Amazon Web Services (AWS) is a **subsidiary of Amazon** providing on-demand cloud computing platforms and APIs to individuals, companies, and governments, on a metered pay-as-you-go basis.
- Amazon Web Services (AWS) is the **world's most comprehensive and broadly adopted cloud platform**, offering over 175 fully-featured services from data centres globally.
- Millions of customers — including the **fastest-growing startups, largest enterprises, and leading government agencies** — are using AWS to lower costs, become more agile, and innovate faster.

How AWS vision came up?

- In 2003, Benjamin Black and Chris Pinkham wrote a paper for Amazon.
- It described “a vision for Amazon infrastructure that was completely standardized, completely automated, and relied extensively on Web Services for things like storage.”
- Black also explained, “Possibility of selling virtual servers as a service.”

History of AWS

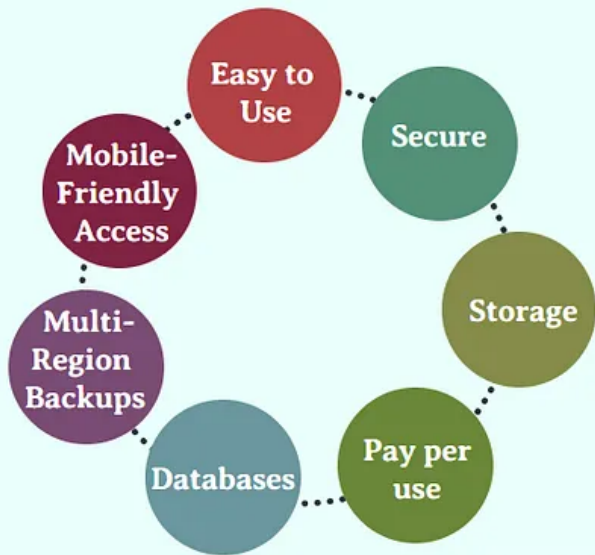
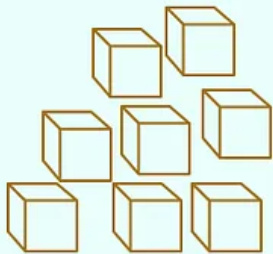
- Amazon launched its first cloud computing service, Simple Storage Service (S3) in March of 2006.
- A popular myth says that Amazon began selling public cloud computing services because it had “excess capacity” from running its e-Commerce website.
- In the early 2000s, Amazon.com's internal development team had a problem.
 - They were adding a lot of software engineers, but despite the growing headcount, the pace of development was staying about the same.
 - **Issue:** Each developer was setting up new and unique compute, storage and database resources for each project.
 - **Solution:** If they could standardize those resources and simplify the process of deploying new IT infrastructure, they might be able to speed things up.

History of AWS ... cont'd

- Amazon began to consider idea of offering different IT services to other companies.
- Idea gained momentum in 2004, Black, Pinkham and their team began work on the project that eventually became AWS.
- After the launch of S3 in the spring of 2006, AWS followed up by taking its Simple Queue Service into production and launching its Elastic Compute Cloud (EC2).
- In 2010, Netflix became the first company to announce publicly that it would run all of its infrastructure on AWS.

Benefits of AWS

Benefits Of AWS



Benefits of AWS

Most functionality

- AWS has significantly more services, and more features within those services, than any other cloud provider – from infrastructure technologies like compute, storage, and databases – to emerging technologies, such as machine learning and artificial intelligence, data lakes and analytics, and Internet of Things.
- This makes it faster, easier, and more cost-effective to move your existing applications to the cloud and build nearly anything you can imagine.
- AWS also has the deepest functionality within those services.
- For example, AWS offers the widest variety of databases that are purpose-built for different types of applications so you can choose the right tool for the job to get the best cost and performance.

Benefits of AWS . . . cont'd

Largest community of customers & partners

- AWS has the largest and most dynamic community, with millions of active customers and tens of thousands of partners globally.
- Customers across virtually every industry and of every size, including **startups, enterprises, and public sector organizations**, are running every imaginable use case on AWS.
- The AWS Partner Network (APN) includes thousands of systems integrators who specialize in AWS services and tens of thousands of independent software vendors (ISVs) who adapt their technology to work on AWS.

Benefits of AWS . . . cont'd

Most secure

- AWS is architected to be the most flexible and secure cloud computing environment available today.
- Its core infrastructure is built to satisfy the security requirements for the military, global banks, and other high-sensitivity organizations.
- This is backed by a **deep set of cloud security tools, with 230 security, compliance, and governance services and features.**
- **AWS supports 90 security standards and compliance certifications**, and all 117 AWS services that store customer data offer the ability to encrypt that data.

Benefits of AWS . . . cont'd

Fastest pace of innovation

- With AWS, you can leverage the latest technologies to experiment and innovate more quickly.
- AWS are continually accelerating pace of innovation to invent entirely new technologies you can use to transform your business.
- For example, in 2014, AWS pioneered the serverless computing space with the launch of AWS Lambda, which lets developers run their code without provisioning or managing servers.
- And AWS built Amazon SageMaker, a fully managed machine learning service that empowers everyday developers and scientists to use machine learning – without any previous experience.

Benefits of AWS . . . cont'd

Most proven operational expertise

- AWS has unmatched experience, maturity, reliability, security, and performance that you can depend upon for your most important applications.
- For over 13 years, AWS has been delivering cloud services to millions of customers around the world running a wide variety of use cases.
- AWS has the most operational experience, at greater scale, of any cloud provider.

Elasticity

- With cloud computing, you don't have to over-provision resources upfront to handle peak levels of business activity in the future.
- Instead, you provision the number of resources that you actually need.
- You can scale these resources up or down to instantly to grow and shrink capacity as your business needs change.

Benefits of AWS . . . cont'd

Agility

- The **cloud gives you easy access to a broad range of technologies** so that you can innovate faster and build nearly anything that you can imagine.
- You can **quickly spin up resources as you need them** - from infrastructure services, such as compute, storage, and databases, to the Internet of Things, machine learning, data lakes and analytics, and much more.
- You can **deploy technology services in a matter of minutes**, and get from idea to implementation several orders of magnitude faster than before.
- This gives you the **freedom to experiment, test new ideas** to differentiate customer experiences and **transform your business**.

Benefits of AWS ... cont'd

Cost Savings

- The cloud allows you to **trade capital expenses** (such as data centres and physical servers) for variable expenses and **only pay for IT as you consume it**.
- Variable expenses are much lower than what you would pay to do it yourself because of the **economies of scale**.
- **Deploy Globally in minutes** - With the cloud, you can expand to new geographic regions and deploy globally in minutes.
- For example, **AWS has infrastructure all over the world**, so you can deploy your application in multiple physical locations with just a few clicks.
- Remember: **Putting applications in closer proximity to end users reduces latency and improves their experience**.

Cloud Services Categories

Snapshot of AWS

The screenshot shows the AWS Management Console interface. At the top, the browser address bar displays the URL `https://us-west-2.console.aws.amazon.com/console/home?region=us-west-2#`. Below the browser, the console header includes the AWS logo, a 'Services' dropdown menu, an 'Edit' button, and a 'Help' dropdown menu. The main content area is titled 'Amazon Web Services' and is organized into several columns of service tiles, each with an icon and a brief description. The services are categorized as follows:

- Compute & Networking:** Direct Connect (Dedicated Network Connection to AWS), EC2 (Virtual Servers in the Cloud), Route 53 (Scalable Domain Name System), VPC (Isolated Cloud Resources).
- Storage & Content Delivery:** CloudFront (Global Content Delivery Network), Glacier (Archive Storage in the Cloud), S3 (Scalable Storage in the Cloud), Storage Gateway (Integrates On-Premises IT Environments with Cloud Storage).
- Database:** DynamoDB (Predictable and Scalable NoSQL Data Store), ElastiCache (In-Memory Cache), RDS (Managed Relational Database Service), Redshift (Managed Petabyte-Scale Data Warehouse Service).
- Deployment & Management:** CloudFormation (Templated AWS Resource Creation), CloudTrail (User Activity and Change Tracking), CloudWatch (Resource and Application Monitoring), Directory Service (Managed Directories in the Cloud), Elastic Beanstalk (AWS Application Container), IAM (Secure AWS Access Control), OpsWorks (DevOps Application Management Service), Trusted Advisor (AWS Cloud Optimization Expert).
- Analytics:** Data Pipeline (Orchestration for Data-Driven Workflows), Elastic MapReduce (Managed Hadoop Framework), Kinesis (Real-time Processing of Streaming Big Data).
- Mobile Services:** Cognito (User Identity and App Data Synchronization), Mobile Analytics (Understand App Usage Data at Scale), SNS (Push Notification Service).
- App Services:** AppStream (Low Latency Application Streaming), CloudSearch (Managed Search Service), Elastic Transcoder (Easy-to-use Scalable Media Transcoding), SES (Email Sending Service), SQS (Message Queue Service), SWF (Workflow Service for Coordinating Application Components).
- Applications:** WorkSpaces (Desktops in the Cloud), Zocalo (Secure Enterprise Storage and Sharing Service).

On the right side of the console, there are three sections:

- Additional Resources:** Includes links for 'Getting Started' (documentation), 'AWS Console Mobile App' (view resources on the go), and 'AWS Marketplace' (find and buy software).
- Service Health:** Shows a green checkmark indicating 'All services operating normally.' It also includes an update timestamp and a link to the 'Service Health Dashboard'.
- Set Start Page:** A dropdown menu currently set to 'Console Home'.

Types of Cloud Services

- Three main types of cloud computing include,
 - Infrastructure as a Service (**IaaS**),
 - Platform as a Service (**PaaS**), and
 - Software as a Service (**SaaS**).
- Each type of cloud computing service provides different levels of control, flexibility, and management so that you can select the right set of services for your needs.

Types of Cloud Services . . . cont'd

Infrastructure as a Service (IaaS)

- IaaS contains the basic building blocks for cloud IT.
- It typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space.
- IaaS gives you the highest level of flexibility and management control over your IT resources.
- It is most similar to the existing IT resources with which many IT departments and developers are familiar.

Types of Cloud Services . . . cont'd

Platform as a Service (PaaS)

- PaaS removes the need for you to manage underlying infrastructure (usually hardware and operating systems), and allows you to focus on the deployment and management of your applications.
- This helps you be more efficient as you don't need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.

Types of Cloud Services . . . cont'd

Software as a Service (SaaS)

- SaaS provides you with a complete product that is run and managed by the service provider.
- In most cases, people referring to SaaS are referring to end-user applications (such as web-based email).
- With a SaaS offering, you don't have to think about how the service is maintained or how the underlying infrastructure is managed.
- You only need to think about how you will use that particular software.

A Software as a Service (SaaS) company is essentially a type of business that delivers software applications over the internet on a subscription basis.

Unlike traditional software companies that require users to purchase and install software on their computers, SaaS companies host and maintain the software centrally, allowing users to access it through a web browser.

SaaS Landscape

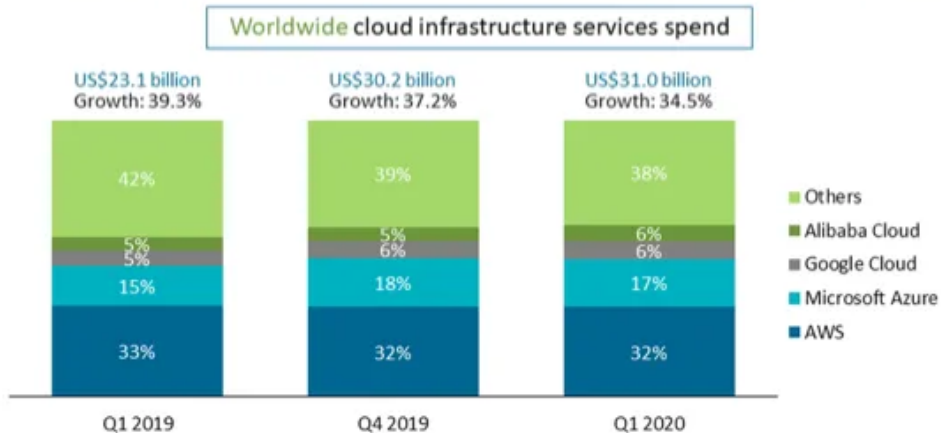
- India is on the cusp of becoming a global SaaS powerhouse.
- SaaS startups in India have been recognized for their innovative solutions and have contributed significantly to the growth of the software industry in the country.
- As B2B SaaS companies in India continue to thrive, they are reshaping the landscape of software development and delivery.
- Their focus on providing scalable and cost-effective solutions has made them attractive options for businesses looking to streamline their operations.
- With the rise of SaaS product companies, India is poised to become one of the biggest players in the global SaaS market, further solidifying its position as a leader in SaaS tech.
- Companies to watch: Zoho, Freshworks, Chargebee, Keka, Whatfix, Atlassian, CleverTap, Facilio, WebEngage, Druva.

Who is using Cloud Services?

- Organizations of every type, size, and industry are using the cloud for a wide variety of use cases, such as data backup, disaster recovery, email, virtual desktops, software development and testing, big data analytics, and customer-facing web applications.
- For example,
 - Healthcare companies are using the cloud to develop more personalized treatments for patients.
 - Financial services companies are using the cloud to power real-time fraud detection and prevention.
 - Video game makers are using the cloud to deliver online games to millions of players around the world.
 - List is long ...

Comparison: Best Public Cloud Providers

Top four providers account for 62% of cloud spend



Source: Canalys estimates, April 2020



Figure: Comparison among Cloud Services Providers ¹

¹<https://www.hava.io/blog/2024-cloud-market-share-analysis-decoding-industry-leaders-and-trends>

Cloud Services Providers - Amazon Web Services (AWS)

- AWS is the undisputed market leader in cloud computing, from overall market share to most expansive cloud offering.
- It has vast resources, allowing it to design and execute new solutions at a dizzying pace, sometimes — or often — faster than customers can understand or incorporate them.
- AWS offers a complete range of IaaS & PaaS services. Among best known are its,
 - Elastic Compute Cloud (EC2),
 - Elastic Beanstalk,
 - Simple Storage Service (S3),
 - Elastic Block Store (EBS),
 - Glacier storage,
 - Relational Database Service (RDS), and
 - DynamoDB NoSQL database.
- AWS offers cloud services related to networking, analytics and machine learning, Internet of Things (IoT), mobile services, development, cloud management, cloud security and more.

Cloud Services Providers - Broad Landscape ... cont'd

Microsoft Azure

- Most market analysts put **Microsoft in number 2 spot behind Amazon.**
- Cloud portfolio is exhaustive:
 - **Azure IaaS & PaaS offerings,**
 - **Microsoft has several SaaS offerings,** including its Office 365 products, the online versions of its Dynamics line of enterprise software and its online developer tools.
- Industry experts believe that Microsoft has **"enterprise IT in its DNA."**
 - **Azure is leading edge** on all the newest cloud technologies, from serverless (FaaS) to microservices to containers.
- It is set up to handle a broad range of clients, from **small medium businesses** (SMB) to a **larger enterprise,** from **multi-cloud to — especially — hybrid cloud.**
- To fill out its offering, it has **longstanding alliances** with other legacy vendors, from **VMware to Oracle.**

Cloud Services Providers - Broad Landscape ... cont'd

IBM Cloud

- Though not considered one of top 3 cloud computing vendors, IBM's cloud business has been coming on strong.
- Particularly advantageous was IBM's acquisition of RedHat, which allows it to leverage RedHat's OpenShift cloud platform, a PaaS solution geared for containers.
- IBM is also strong in hybrid — befitting its customer base of large enterprise clients, which in some cases are working on a classic lift and shift migration to the cloud, for which IBM is well suited.
- One of IBM's key cloud service is its Bluemix PaaS, which is aimed primarily at enterprise development teams.
- IBM also provides a lot of enterprise software on a SaaS basis, and it sells cloud infrastructure, cloud management tools and cloud managed services.

Cloud Services Providers - Broad Landscape ... cont'd

Google Cloud Platform (GCP)

- Ranked 3rd behind AWS and Azure, Google is a major contender in cloud market.
- Its deep strength data analytics and artificial intelligence is only growing, and will likely be a major force in years ahead.
- Its industry-leading creations with TensorFlow and Kubernetes are examples of sophisticated solutions that earn GCP leader status in the cloud market.
- Like AWS and Azure, GCP offers an extensive range of IaaS and PaaS services that span computes, storage, networking, big data, machine learning, developer tools and security.
- GCP best-known cloud offerings include,
 - Compute Engine,
 - App Engine,
 - Container Engine,
 - Cloud Storage and BigQuery.

Cloud Services Providers - Broad Landscape ... cont'd

Oracle Cloud Infrastructure (OCI)

- OCI is generally favoured by larger enterprise and existing Oracle customers, as opposed to SMB and developers, two cohorts that have fueled AWS's rise.
- **Strength:** Hybrid cloud computing
OCI is a good fit for larger customers that have a legacy investment in on-premise IT infrastructure and data centres.
- Company's **Oracle Dedicated Region Cloud Customer solution** is an on-prem offering that is a well designed, technologically advanced mirror of cloud version.
- **Important Tie-up:** Oracle has a partnership with Microsoft Azure for those cloud functionalities it can't currently support.

Cloud Services Providers - Broad Landscape ... cont'd

Alibaba Cloud

- Leader in the Asian cloud market, Singapore-based Alibaba is forecast to grow rapidly as cloud computing continues to grow in a variety of Asian countries.
- It benefits from its deep strength in e-Commerce, which is Alibaba's core business.
- With this deep expertise, it enables customers to access new sources of Internet-based income.
- The company offers a cloud-based AI service, which will almost certainly be a major growth area in the near term future.
- Well regarded for its data solutions, providing interoperability with myriad other data platforms. Major strength for its e-commerce clients.
- Alibaba is currently largely an Asian player, it's entirely possible that it will expand into other geographic markets as its cloud and data offerings mature.

AWS Use Cases

AWS Use Cases

Millions of customers — including the fastest-growing startups, largest enterprises, and leading government agencies — are using AWS to lower costs, become more agile, and innovate faster.

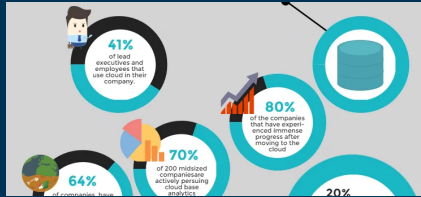
Below are few areas and selected top companies that have been using AWS.

- **Aerospace** (Maxar, ESA etc.)
- **Gaming** (MPL, FanFight, Gammation etc.)
- **Education** (Coursera, BYJU's etc.)
- **Telecommunication** (Pinterest, Vodafone, Aircel etc.)
- **Entertainment** (Netflix, Hotstar etc.)
- **Media** (BBC, The Hindu, Punjab Kesri etc.)
- **Software** (Share chat, Slack etc.)

Challenges in Use of AWS

- **AWS weakness:** Lack of offerings for hybrid cloud deployments.
- Analysts say that most enterprises will be pursuing a hybrid cloud, multi-cloud strategy, and **AWS competitors Azure and IBM have an advantage in this area.**
- Because **many large organizations already use Microsoft and IBM products in their data centres**, they naturally gravitate to these other providers for the public cloud portion of their hybrid clouds.
- **Watch AWS:** Will it be the best option for emerging technologies like AI, ML, IoT and containerized deployments.
- **All leading vendors are competing heavily in these areas, and AWS will have to continue to innovate if it wants to retain its position as the market leader.**
- In the technology industry, markets can shift very quickly, and being the number one provider today is no guarantee of future performance.

Cloud Computing Facts



- Cloud is being widely used and that people spend a great deal of money creating, managing, and upgrading their cloud computing systems.
- Some fascinating facts around:
 - Nearly one-half of US Government agencies use cloud services.
 - Banking produces the most activity within the cloud.
 - Cloud computing market is projected to reach \$250 billion by 2026.
 - 60% of U.S. IT decision-makers trust the security of the cloud.
 - Within the next five years, the public cloud will experience a 100% growth.
 - 80% of companies that adopt cloud see improvements within their IT departments within six months.

Related Case-Studies - Netflix, Siemens, Adobe, Hotstar, BMW

Netflix AWS Casestudy

- Netflix was originally a DVD shipping business where they would send out DVDs of your chosen programs to you.
- Reason for cloud transition
 - Going well until 2008 - a major database loss and for 3 days could not ship out any DVDs to their customers.
 - Netflix realized that they had to shift from continuous vertical scaling which leads to single points of failure to a more reliable and scalable horizontal scaling system.
- Netflix chose AWS despite having Amazon as a competitor (Amazon has their own streaming service known as Amazon Prime) because AWS provided them with the greatest scaling capabilities and the biggest set of available features.
- Time taken for cloud migration: 7 years
(Netflix to shut down all data centres and moving completely to cloud).
- Business Benefits: Moving to the cloud has allowed Netflix to keep its existing members well engaged with overall viewing growing exponentially.

Netflix AWS Casestudy . . . cont'd

- Netflix itself has continued to evolve rapidly by using many new features and relying on ever-growing volumes of data.
- What if not AWS?
 - Supporting fast growth would not be possible using in-house data centres.
 - Netflix could not have racked the servers fast enough to support their own growth. Cloud brings elasticity, which allows adding thousands of virtual servers & petabytes of storage within minutes without the need to maintain large capacity machines.
 - Rapid Expansion was not possible
From 2008 to 2016, Netflix has expanded into 130 new countries.
 - Use of multiple AWS Cloud regions spread all over the world to create a better and more enjoyable streaming experience for Netflix members wherever they are.
- Netflix relies on AWS for
 - All its scalability,
 - computing and storage needs (not only video streaming) — Netflix business logic, distributed databases, big data processing, analytics, recommendations, transcoding and other functions go through their Cloud infrastructure.

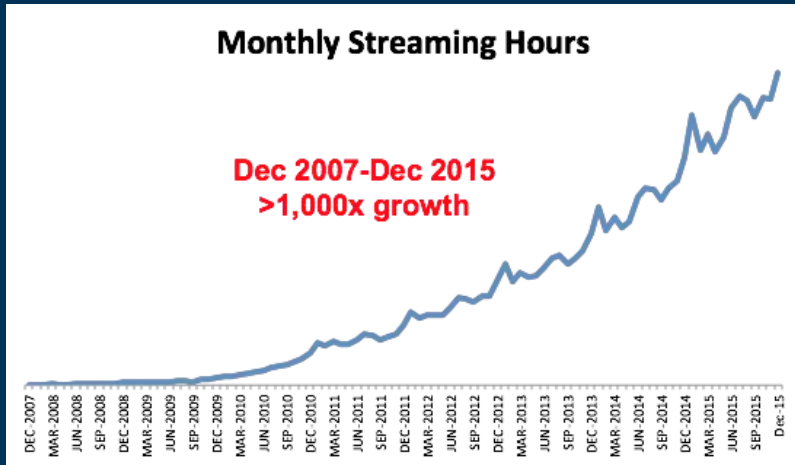
Netflix AWS Casestudy ... cont'd

- Netflix also has its own Content Delivery Network (CDN) known as Netflix Open Connect which is used to deliver videos globally in an efficient manner.
- What issues Netflix faced before moving to AWS?
When using their own data centres, they faced a lot of outages.
- Realty facts
 - Cloud Computing is not perfect either, though Netflix has hit some rough patches in the cloud, a steady increase in the overall availability has been noticed.
 - Failures are ultimately unavoidable in any large-scale distribution system
Cloud-based system allows creating redundancy which becomes quite helpful.
 - Cloud Computing has made it possible to survive failures without impacting the member experience.

Netflix AWS Casestudy ... cont'd

- Economies of Scale helps Netflix in this scenario - However, Netflix did not shift to cloud for cost reduction reasons.
- Time to fully migrate to AWS: 07 years.
- Challenges in Migration
 - Moving to the cloud is a lot of work and a lot of factors need to be considered.
 - Netflix could easily move all of its existing systems to AWS but bringing existing systems also brings all problems / limitations that were present.
 - Netflix took the cloud-native approach
Rebuilt all of their technology and fundamentally changed the way they operate the whole company.
 - Netflix migrated from a single application to thousands of micro-services.

Netflix AWS Casestudy ... cont'd



Siemens AWS Casestudy

■ About Siemens

Global leader in electrification, automation, and digitization, has driven innovation across industries for nearly 175 years.

■ Siemens uses AWS services to carry out digital transformation —

- Bringing IIOT to railways and factories, developing intelligent infrastructure for buildings and distributed energy systems, implementing AI into its cybersecurity platform, and more.
- To realize value across the entire business and embrace Industry 4.0, **Siemens built MindSphere**, its open Internet of Things (IoT) operating platform hosted on AWS.
- **MindSphere on AWS**, used to achieve world-class levels of manufacturing efficiency.

■ AWS Advantage

Driven by automation, IoT, and cloud computing, Siemens can now solve business problems with the data that it collects, analyzes, and monitors.

Adobe AWS Casestudy

■ About Adobe Systems

- Delivers enterprise software to large multi-national corps., & govt. agencies.
- Eg. Apps. - Adobe LiveCycle Forms, Adobe Connect conferencing software, Adobe CQ5, Adobe Premiere Pro, Adobe After Effects, Adobe Photoshop, & Adobe Media Encoder

■ How Adobe uses AWS?

- Adobe uses AWS to provide multi-terabyte operating environments for its customers.

■ AWS Advantage

- Adobe can focus on deploying and operating its own software instead of infrastructure.
- Adobe Creative Cloud Suite can now run on AWS, enabling customers to unlock creativity from anywhere in the world.
- With the joint effort between Adobe and AWS, applications are no longer bound to your local desktop.

Adobe AWS Casestudy . . . cont'd

- Customers frequently ask if they can run their editing software in AWS - patterns of use can be described in the form of distinct use cases, or what we call **user personas**.
- Some of the most common personas are **News/Sports, Creative, and Promo**.
- Personas have different demands in terms of storage, network bandwidth, disk I/O, CPU, and memory.
Some workflows are a challenge in cloud, including colour grading, colour fidelity, & multi-channel audio support.
- As Virtual desktop infrastructure (VDI) protocols evolve to support functionality such as 10-bit colour and more audio channels, these workflows can be enabled, and AWS has designed its templates with flexibility in mind to accommodate future feature improvements.

Adobe AWS Casestudy . . . cont'd

- **Note:** Minimizing network latency has direct impact on workstation interactivity & that being physically closer to an AWS region provides better user experience.
- **Locations such as Los Angeles have dedicated AWS Local Zones**
Reduces latency by placing AWS compute, storage, database, and other select services closer to a large population, industry, and IT centres.
- **AWS Advantage**
 - When operating in a production facility, studio, or creative office, AWS Direct Connect enhances connectivity through dedicated capacity,
 - Adds a layer of security abstracting the public internet path.
- When choosing a deployment location or region, a target latency of around 30ms or less will provide an optimal experience.
- **Higher latency may lead to a lag in peripheral device interactivity** such as general playback and graphical interface activities.

Hotstar AWS Casestudy

- **About Novi Digital**

India's largest media companies, providing streaming video for more than 40 channels in seven languages to more than 720 million users.

- **AWS for HotStar**

To provide cost-effective scalability and reliability for its OTT platform.

- AWS provides hotstar with 5700 Gbps of bandwidth during a live match.

- **India Context**

Backend architecture by AWS - Hotstar is a popular choice over Netflix and Amazon prime video, even having the toughest competition.

- **Advantage Hotstar:**

Using AWS, Novi Digital drastically reduced both cost and time to market for its over-the-top (OTT) content platform, hotstar.com.

BMW AWS Casestudy

■ About BMW Group

Leading manufacturers of premium cars and mobility services in the world, with brands such as Rolls Royce, BMW, and Mini.

- BMW is using **AWS for its connected-car app** that collects sensor data from **BMW 7 Series cars** to give drivers dynamically updated map information.
- BMW built its **car-as-a-sensor (CARASSO) service** in only six months leveraging,
 - Amazon Simple Storage Service (Amazon S3),
 - Amazon Simple Queue Service (Amazon SQS),
 - Amazon DynamoDB,
 - Amazon Relational Database Service (Amazon RDS), and
 - AWS Elastic Beanstalk.
- **AWS Advantage:** CARASSO can adapt to rapidly changing load requirements that can scale up / down.
- **Scale:** In 2018, CARASSO processed data collected by a fleet of 100,000 vehicles travelling more than eight billion kilometres.

Thank You